CoalGuard OS — Tech Stack & Architecture

Smart India Hackathon 2026 • Ministry of Coal / DGMS Solution

🎯 1. Executive Summary & Problem Scope

CoalGuard OS is an AI-powered Smart Governance & Statutory Compliance Monitoring Platform engineered specifically for Indian coalfields (Jharia, Bokaro, Korba, Raniganj).

Key Real-World Mine Constraints Solved:

- **Underground Zero-Connectivity:** 300m deep seams have zero internet; system features full offline local queueing with Base64 media caching and automatic surface reconnect synchronization.
- **Glove-Friendly Worker Dictation:** Native Web Speech API allows miners wearing heavy gear to dictate in Hindi or English.
- **Autonomous IoT Sensor Ingestion:** Continuous 3-second multi-gas monitoring (Methane CH_4, CO, Airflow, Strata Pressure) triggers instant zero-human hazard triage when statutory limits are crossed.
- **DGMS CMR 2017 Compliance:** Gemini AI structures raw field logs into statutory incident reports with immediate 3-step containment SOPs.

🏛️ 2. High-Level Architecture Diagram

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|                                  COALGUARD OS SYSTEM ARCHITECTURE                             |
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 [ FIELD LEVEL: Ground Miners / Inspectors ]      [ COMMAND LEVEL: DGMS General Managers / Audit ]
    |                                                        |
    |-- 🎤 Native Hindi/English Voice Dictation              |-- 📊 Real-Time Analytics & Recharts
    |-- 🔴 Underground Offline-First Queue                   |-- 📡 Live 3s IoT Telemetry Transducers
    |-- 📍 Geo-Tagged GPS Pit Location                      |-- 👷 Contractor Compliance Scorecard
    |-- 📎 Multi-Evidence Photos / Gas Logs                  |-- 🚜 Fleet Maintenance Lifecycle Predic
    \___________________________                            |-- 🔮 7-Day Predictive Risk Surge AI
                                \                          ___________________________/
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                      +-----------------------------------+
                      |  REACT 19 + VITE SINGLE PAGE APP |
                      |  - React Router Dom v7, PWA      |
                      |  - Bilingual i18n (Hindi/English)|
                      +-----------------------------------+
                                     |         ^
                     REST / Multipart |         | WebSockets (Socket.io)
                     Form Requests    v         | (Real-time Live Chat & Radar)
                      +-----------------------------------+
                      | NODE.JS + EXPRESS.JS API SERVER  |
                      | - JWT RBAC Auth & Multer Parsing |
                      | - Statutory CSV Audit Generator  |
                      +-----------------------------------+
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                 +-------------+ +-------------+ +--------------------+
                 |   MONGODB   | | CLOUDINARY  | | GOOGLE GEMINI 1.5  |
                 |   ATLAS     | |  MEDIA CDN  | |   FLASH AI ENGINE  |
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```

💻 3. Complete Technology Stack Matrix

Frontend (Client-Side)

TECHNOLOGY	VERSION	CORE ROLE IN COALGUARD OS
React	v19.0.0	Modern concurrent component-based UI rendering and state management.
Vite	v8.1.5	Ultra-fast ES-module build tooling, HMR, and LightningCSS minification.
React Router Dom	v7.1.5	Client-side routing between Landing, Field Portal, and Admin Command Center.
Recharts	v2.15.1	Interactive SVG charts for DGMS category distributions and 7-day velocity.
Web Speech API	Native API	Hindi/English speech recognition for ground miners wearing dust masks/gloves.
Geolocation API	Native API	Satellite GPS latitude/longitude acquisition and Google Maps linking.
PWA / Service Worker	Native Web	Installable progressive mobile application for field inspectors.
Offline Sync Queue	localStorage	Zero-connectivity underground report caching and auto-sync on reconnect.

Backend, Database & AI Pipeline

TECHNOLOGY	VERSION	CORE ROLE IN COALGUARD OS
Node.js	LTS	Asynchronous, event-driven JavaScript backend runtime.
Express.js	v5.2.1	RESTful microservice routing for incidents, analytics, and auth.

TECHNOLOGY	VERSION	CORE ROLE IN COALGUARD OS
Socket.io	v4.8.3	Full-duplex real-time WebSocket communication for live control room threads.
MongoDB Atlas	Cloud Cluster	High-availability NoSQL document database storing geo-tagged mine logs.
Mongoose ODM	v9.7.4	Data schemas, enum constraints, auto-indexing, and query aggregation.
Google Gemini 1.5 Flash	@google/genai	Autonomous DGMS hazard triage, severity scoring, and 3-step containment SOPs.
Cloudinary CDN	v2.10.0	Cloud asset storage and WebP compression for heavy inspection photos.
JWT & Bcrypt.js	Security	Role-Based Access Control (Admin vs Inspector) and salted password hashing.

⚙️ 4. Key Functional Modules

📡 1. Live IoT Telemetry

Continuous 3s telemetry for Methane, CO, Airflow, and Strata Pressure with 1-click statutory breach simulation.

👷 2. Contractor Intel

Automated compliance scorecards (BGR, VPR, Thriveni) evaluating worker muster, violations, and contract audit warnings.

🚜 3. Fleet Maintenance

Machinery lifecycle tracker (Dumpers, Shovels, Ventilation Fans) with 15-day advance AI overhaul alerts.

🔮 4. Predictive Surge AI

7-Day in-memory risk forecaster projecting upcoming hazard spikes and staffing allocations.

🎓 5. Team Viva & Pitch Study Guide

Q1: How does Offline Sync operate in zero-network underground seams?

When `navigator.onLine` is false, the submission handler serializes report data and attached photos (Base64) into `localStorage` under `coalguard_offline_queue`. When network connectivity is restored at the surface, `window.addEventListener('online')` automatically dispatches queued items to the server with zero data loss.

Q2: How is Google Gemini 1.5 Flash integrated with DGMS statutes?

The backend streams raw miner observations to `@google/genai` with a specialized prompt acting as a DGMS Chief Safety Auditor. Gemini classifies issues into statutory categories (Safety, Environment, Equipment, Labour, Production, General) according to **CMR 2017 Regulation 153** and generates a 3-step immediate containment SOP.

Q3: Why was Socket.io chosen over standard polling?

Mining emergencies (like explosive gas spikes) require sub-second notification. Socket.io provides bidirectional WebSocket event streaming, instantly pushing live incident alerts, status updates, and control room resolution chats without client polling overhead.